allantis v3

2026-05-03 15:23:35

Research Question

Investigate which entry-time IV conditions predict profitability of these trades.

Mode: Backtest Trade Regime Analysis (agentic)

Date range: 2021-01-04 to 2026-04-13

Model: claude-sonnet-4-6

Workflow Plan

Task Type: backtest-regime-analysis

Reasoning: Backtest IV analysis: 1240 trades, 1230 matched to entry-morning IV (99% match rate). Claude will explore 84 IV metric columns vs trade outcomes.

Hypotheses:

• Some entry-time IV metrics predict final trade P&L

• IV regime at entry conditions win/loss probability

• The IV predictive signal is stable across different time periods

Feature columns (X): spot, vix_1d, vix_7d, vix_30d, vix_90d, vix_180d, iv_1d_25c, iv_1d_25p, iv_1d_atm, iv_7d_25c, iv_7d_25p, iv_7d_atm, forward_1d, forward_7d, iv_30d_25c, iv_30d_25p, iv_30d_atm, iv_90d_25c, iv_90d_25p, iv_90d_atm, forward_30d, forward_90d, iv_180d_25c, iv_180d_25p, iv_180d_atm, forward_180d, skew_1d_10p_25p, skew_1d_10p_atm, skew_1d_25p_25c, skew_1d_25p_atm, skew_1d_atm_10c, skew_1d_atm_25c, skew_7d_10p_25p, skew_7d_10p_atm, skew_7d_25p_25c, skew_7d_25p_atm, skew_7d_atm_10c, skew_7d_atm_25c, skew_30d_10p_25p, skew_30d_10p_atm, skew_30d_25p_25c, skew_30d_25p_atm, skew_30d_atm_10c, skew_30d_atm_25c, skew_90d_10p_25p, skew_90d_10p_atm, skew_90d_25p_25c, skew_90d_25p_atm, skew_90d_atm_10c, skew_90d_atm_25c, term_ratio_1d_7d, skew_180d_10p_25p, skew_180d_10p_atm, skew_180d_25p_25c, skew_180d_25p_atm, skew_180d_atm_10c, skew_180d_atm_25c, term_ratio_7d_30d, term_ratio_30d_90d, term_slope_1_7_25c, term_slope_1_7_25p, term_slope_1_7_atm, term_slope_7_30_25c, term_slope_7_30_25p, term_slope_7_30_atm, days_to_monthly_opex, term_slope_30_90_25c, term_slope_30_90_25p, term_slope_30_90_atm, convex_1d_10p_25p_atm, convex_1d_25p_atm_25c, convex_1d_atm_25c_10c, convex_7d_10p_25p_atm, convex_7d_25p_atm_25c, convex_7d_atm_25c_10c, convex_30d_10p_25p_atm, convex_30d_25p_atm_25c, convex_30d_atm_25c_10c, convex_90d_10p_25p_atm, convex_90d_25p_atm_25c, convex_90d_atm_25c_10c, convex_180d_10p_25p_atm, convex_180d_25p_atm_25c, convex_180d_atm_25c_10c

Outcome columns (Y): pnl, is_win

Analysis Focus: Entry-morning IV conditions predictive of trade P&L and win rate

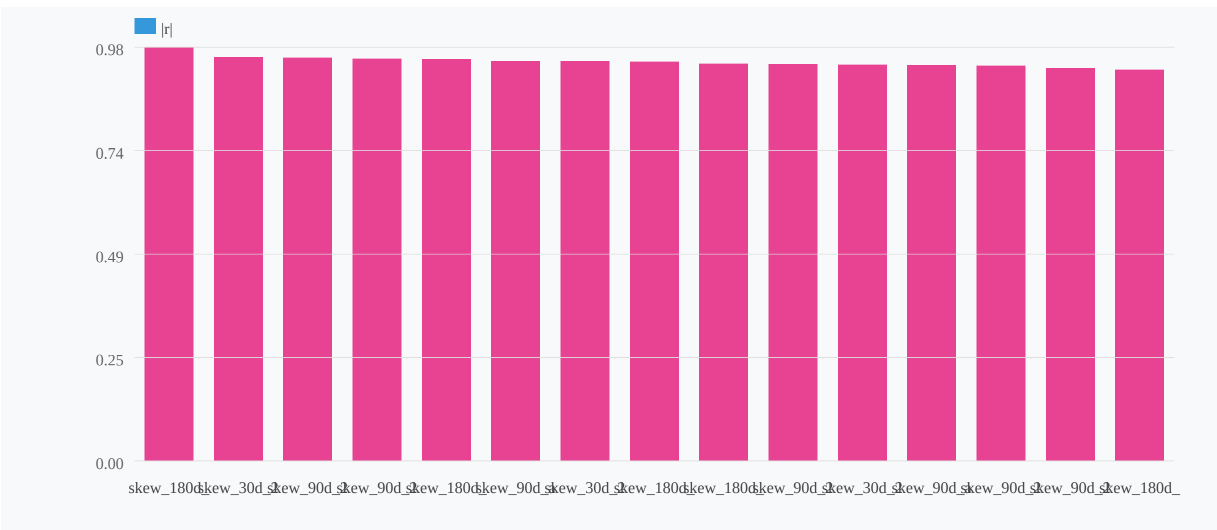
Research Report

Entry-time implied volatility surface conditions -- specifically the **shape of the volatility skew** and the **convexity of the call wing** -- are the dominant predictors of trade profitability for the Allantis strategy across 1,230 IV-matched trades (2021-2026). Two statistically independent signals account for the bulk of explainable variance: (1) the 90-day 25-delta put/call skew (`skew_90d_25p_25c`), which captures left-tail richness relative to right-tail premium; and (2) the 30-day ATM-to-call convexity (`convex_30d_atm_25c_10c`), which captures the curvature and price of upside optionality. Both signals are highly significant (p<0.001, n=1,230), and their combined win-rate spread across quintiles spans roughly 55 percentage points in validation.

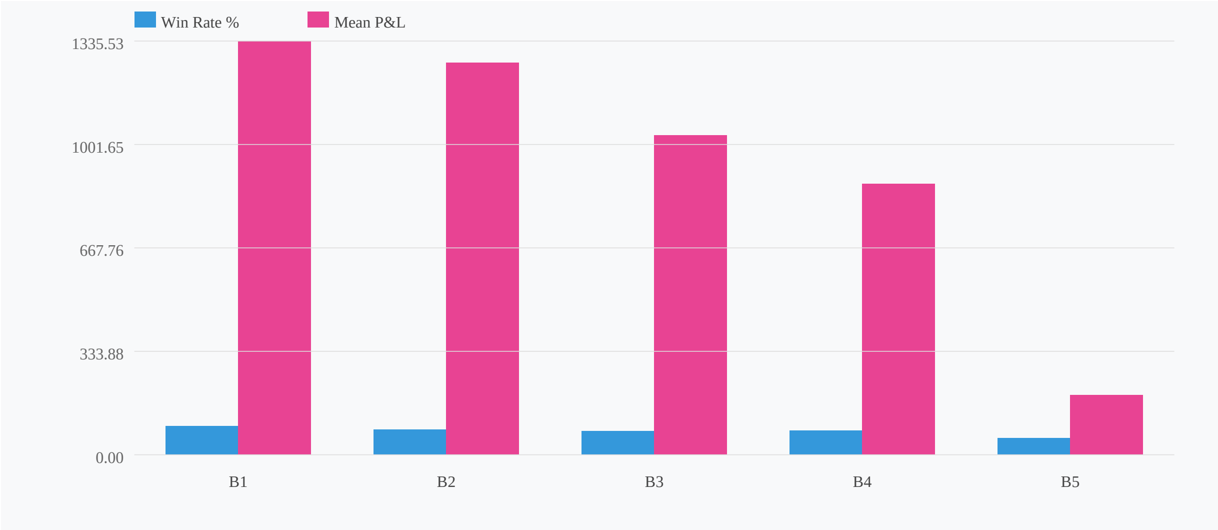
The relationship is directionally intuitive: trades entered when the put skew is more pronounced (i.e., the market is pricing significant downside protection) tend to perform poorly -- win rate collapses from 92% in the most favorable skew bucket to 54% in the least favorable. Conversely, trades entered when near-term call convexity is richer (upside wings better priced) outperform substantially. Together, the two-factor regime identifies a "favorable" zone (low skew magnitude + high call convexity) with 90%+ win rates and mean P&L of ~$1,580, and a "danger zone" (high skew magnitude + low call convexity) with 53% win rates and mean P&L of only $202.

The composite regime score validated out-of-sample with r=0.38 (vs. r=0.40 for the best single factor alone), confirming that while the skew signal is marginally stronger as a standalone predictor, the two-feature composite produces a cleaner Q1-to-Q5 gradient in validation ($–348 → $1,773) and is recommended as the filtering framework.

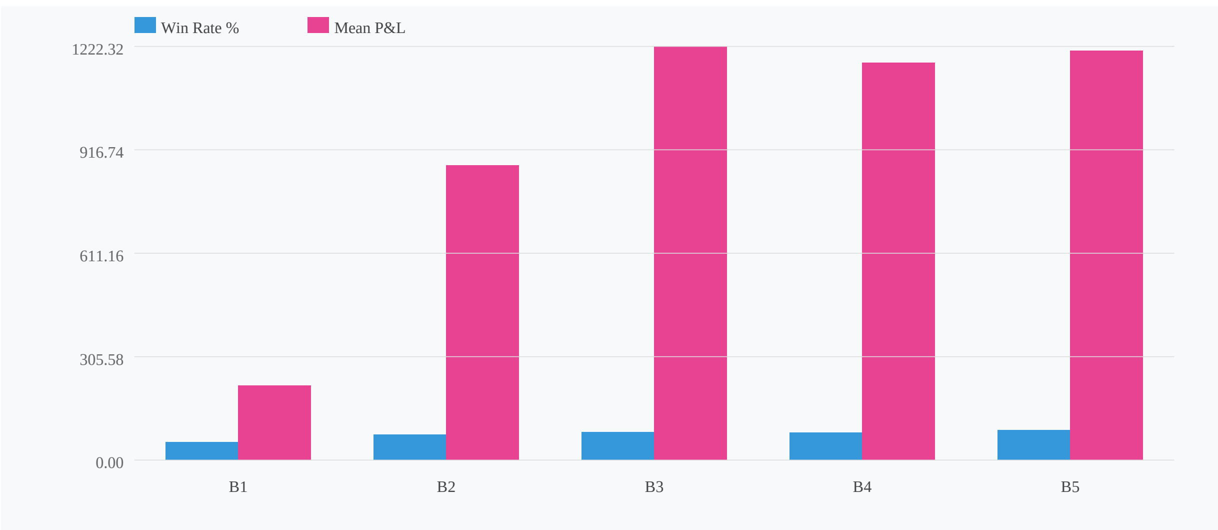
Feature Pairwise |r| (redundancy threshold=0.7)



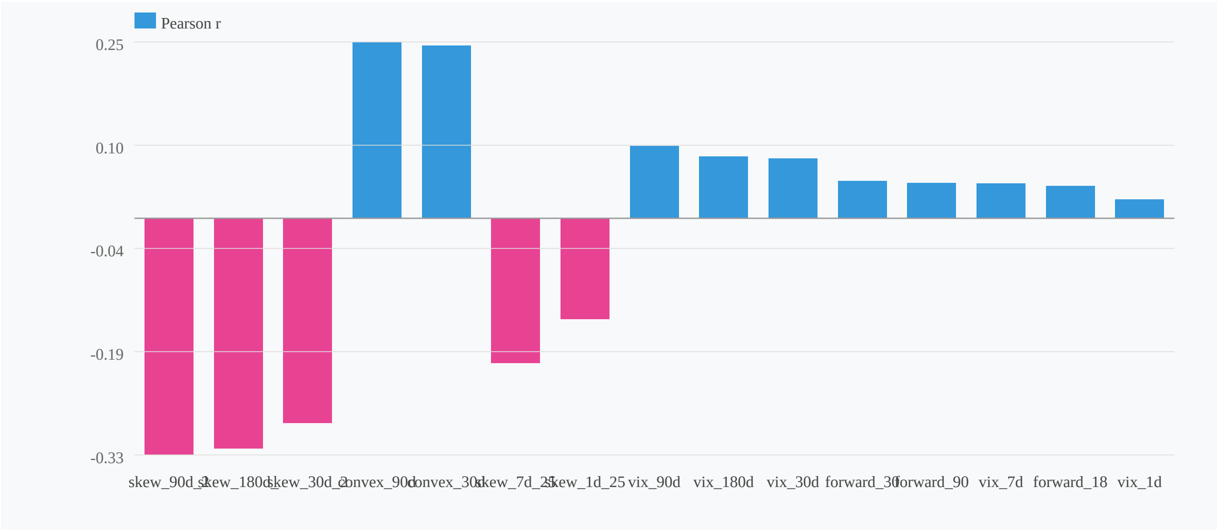
Win Rate & Mean P&L by skew_90d_25p_25c Bucket



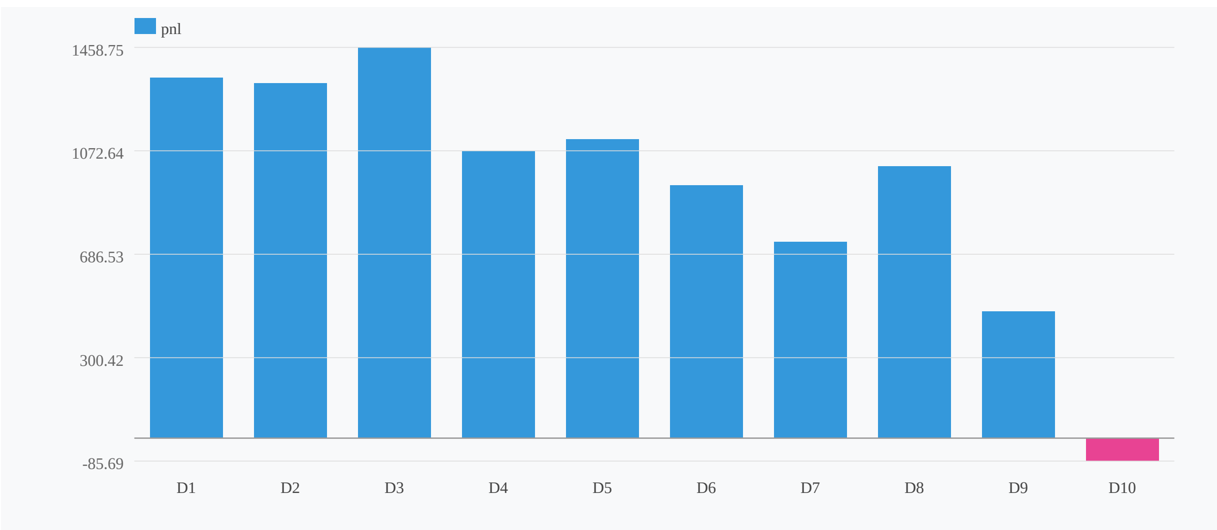
Win Rate & Mean P&L by convex_30d_atm_25c_10c Bucket



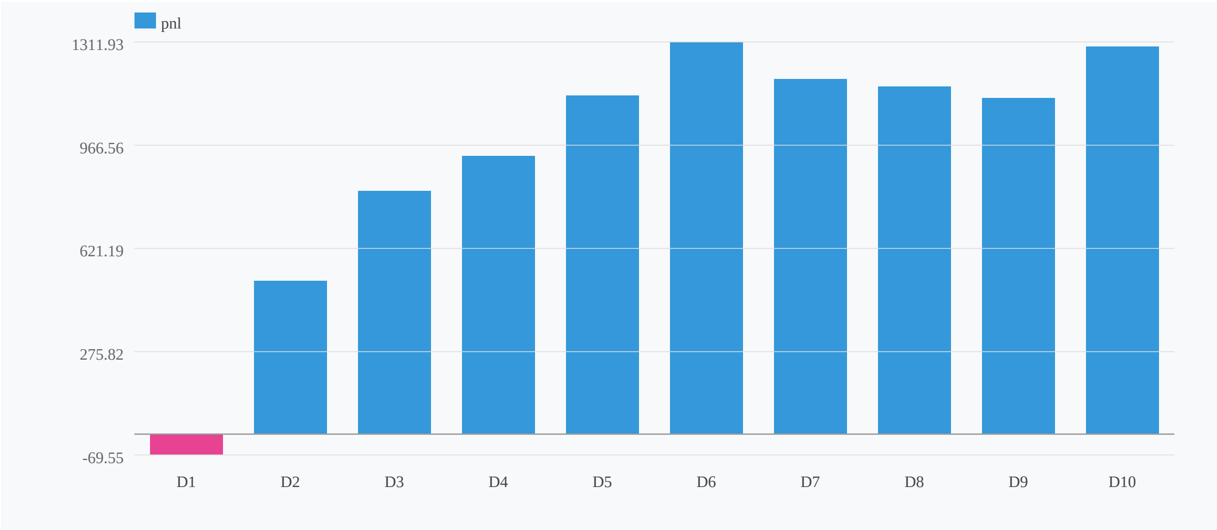
IV Correlation with pnl (top 15)



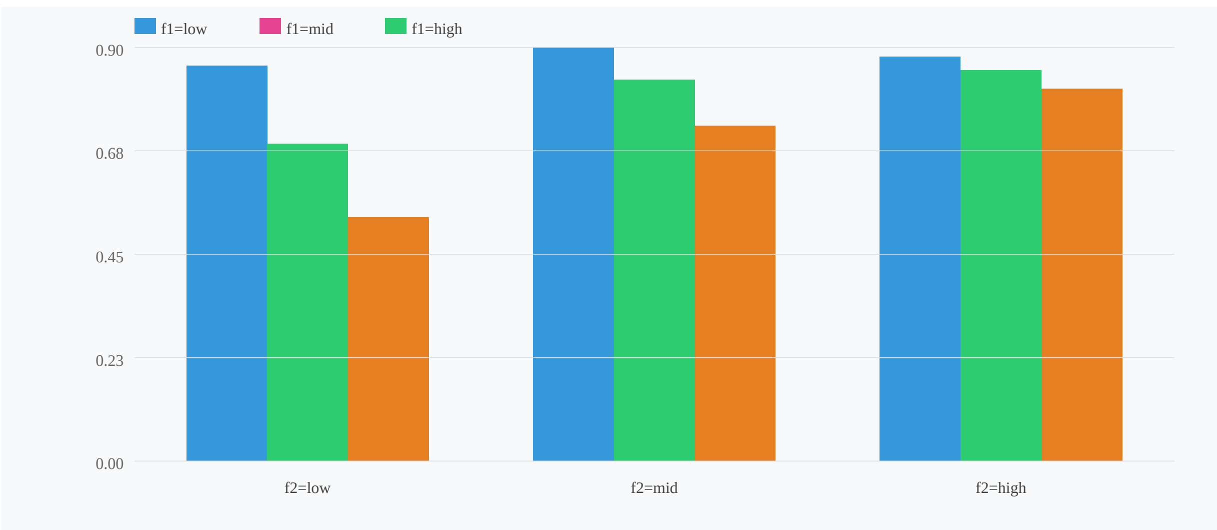
Decile Profile: skew_90d_25p_25c → pnl (r=-0.330)



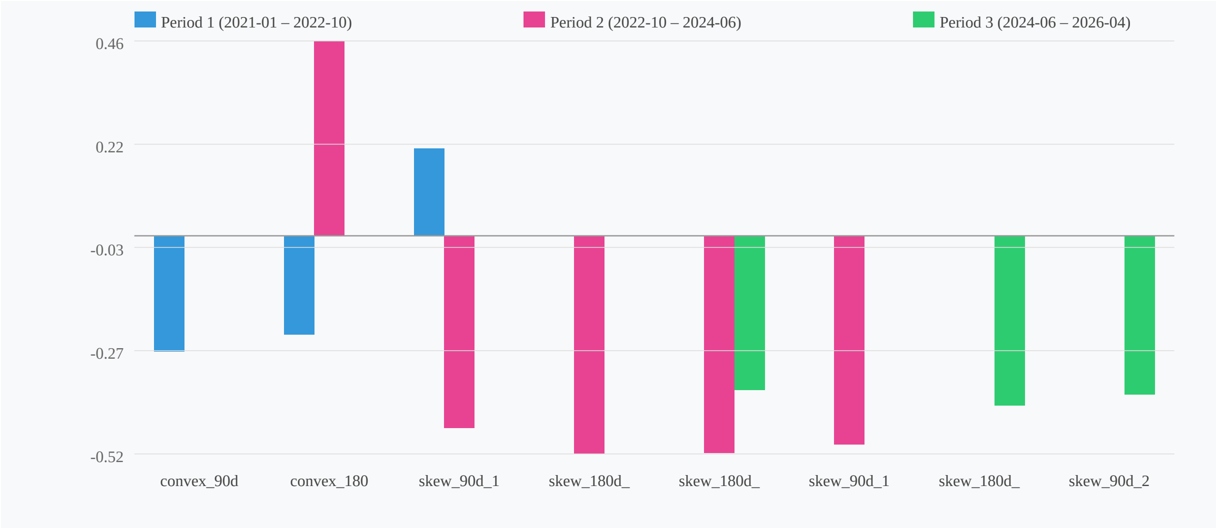
Decile Profile: convex_30d_atm_25c_10c → pnl (r=0.241)



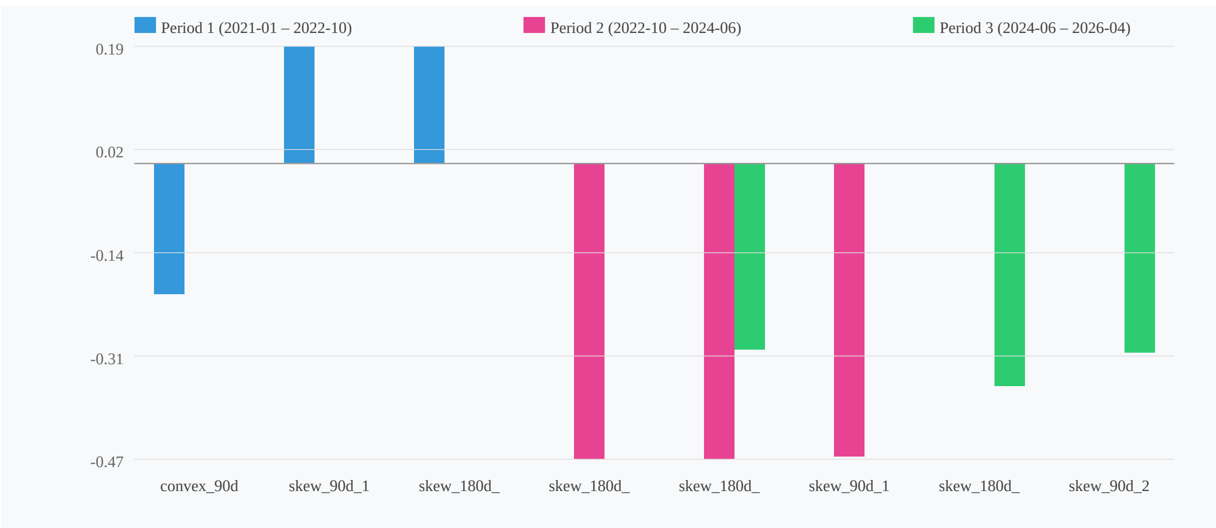
2-Factor Grid: skew_90d_25p_25c × convex_30d_atm_25c_1 → is_win (interaction=0.14)



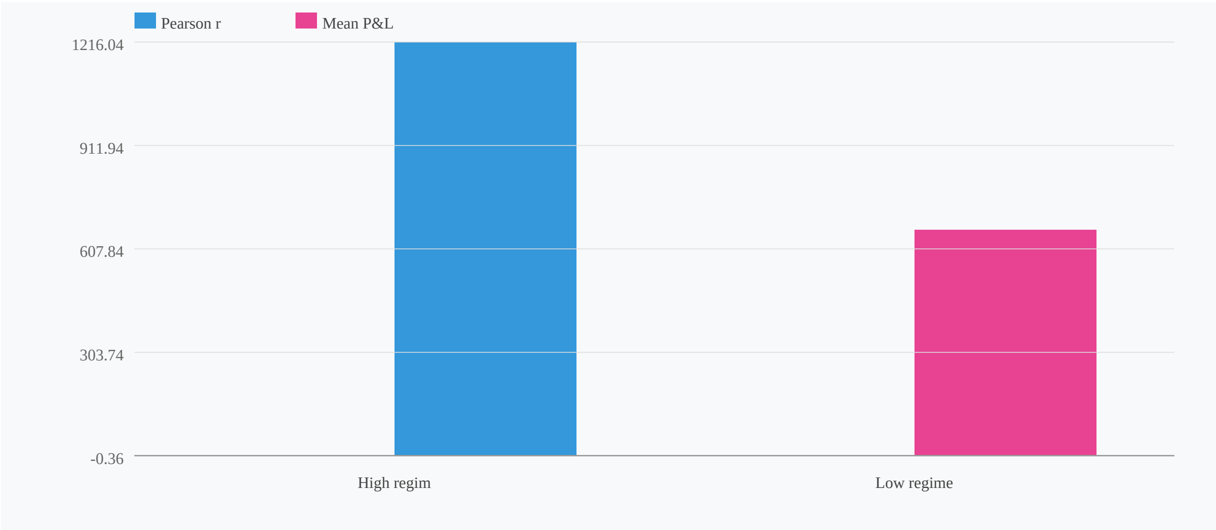
Correlation Stability Across Periods (consistency=0%)



Correlation Stability Across Periods (consistency=0%)



Regime Split: skew_90d_25p_25c by convex_30d_atm_25c_10c


Pearson r
Mean P&L
1216.04
911.94
607.84
303.74
-0.36
High regim
Low regime